# When did plants first live on land? What the rocks can and cannot tell us

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## The puzzle

Every forest, grassland and crop descends from a few small plants that left the water hundreds of millions of years ago. Land plants changed the air, the soils and the climate, and built the stage on which animals came ashore. Yet scientists disagree by tens of millions of years about when the plants we know today began to diversify.

One way to date that is a *molecular clock*: compare the DNA of living plants and work out how long their lineages have been diverging. What it dates is the common ancestor of the living species, which need not be the first plant on land. Clocks need fossils to set the time scale, and they lean heavily on one kind of fossil argument: a *maximum*. A maximum says "if this group had existed before a certain date, we would have found its fossils in older rocks, and we have not". Change the maximum and the clock's answer changes with it.

Three recent clock studies show how much rides on this. Two place the common ancestor of all living land plants 515–474 million years ago (Ma), in the Cambrian or earliest Ordovician. The newest, using a different set of fossil calibrations, places it at 457–450 Ma (Morris et al. 2018; Harris et al. 2022; Hooper et al. 2026). For the ancestor of living *vascular* plants, the group with internal plumbing that includes ferns, conifers and flowering plants, estimates run from about 452 to 428 Ma (figure below).

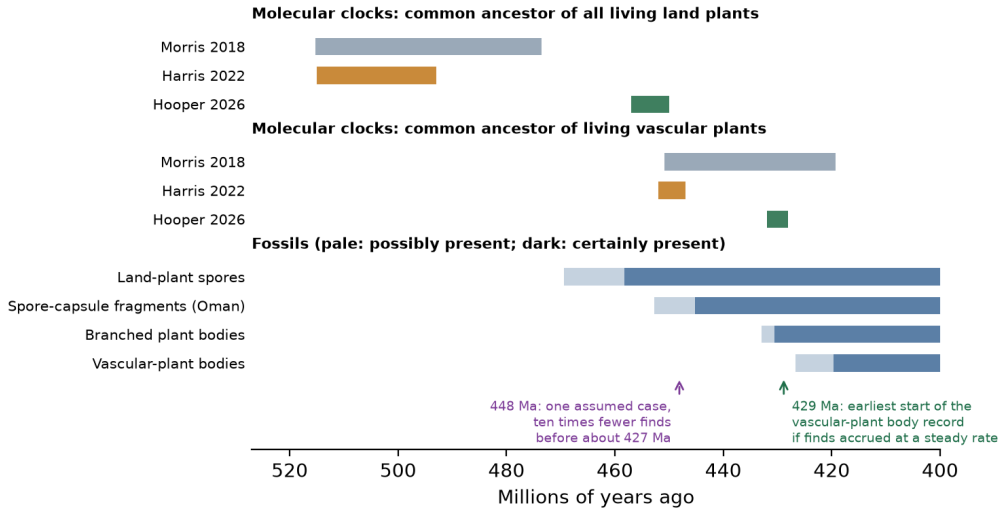



*Figure. Molecular-clock estimates for two key common ancestors, set against what the rocks show. Arrows: how far back the record of vascular-plant bodies could reach if finds accrued at a steady rate (green), and in one assumed case with ten times fewer finds before about 427 Ma (purple); the fossils alone cannot choose between them.*

## Why the rocks are hard to read

The oldest traces of land plants are spores: tough, microscopic, and made in huge numbers. They appear in rocks around 460 Ma. Fragments of spore capsules follow within about ten million years, and branched plants, with stems and capsules often only millimetres long, appear about 30 million years after the spores. Early land plants were small, lacked wood, may have been rare or local, and rotted easily. So the absence of plant fossils from older rocks might mean the plants were not there, or only that they were not preserved, found and recognised. A maximum has to assume the first.

## What we did

We asked a simple question: for each of the three maxima that divide the clocks (515.5, 459.1 and 431.2 Ma), what can the actual fossil record test? We used three bodies of evidence.

- **Spores from Libyan boreholes.** In one North African basin, 28 rock samples older than about 443 Ma contain no spores bearing the three-rayed mark of more advanced land plants, while 51 of 59 younger samples do. Only 10 of the older samples may be older than 459 Ma, and those are broadly dated. It is the only place we found where "not found in older rock" can be tested with numbers.
- **Spores from a Saudi Arabian borehole**, as a rougher check.
- **The dated fossils of early plant bodies** that we could link to a place and an age: 77 dated places, 50 of them with members of the vascular-plant group, plus one set of spore-capsule fragments from Oman. From how these finds are spread through time, a standard method sets a limit on how far back a group's fossil record could reach. The limit depends on how often the plants left fossils that were later found and recognised, so we tried several assumptions.

## What we found

1. **The oldest maximum (515.5 Ma) rests on an argument the rocks we assessed cannot put numbers on.** Its authors argue that older marine rocks preserve tough algal cysts but no land-plant spores, so the spores' absence is real. Reasonable, but whether land spores reached those seas, and would have been found, cannot be measured here.
2. **The 459.1 Ma maximum gets one real test, and it is local and conditional.** In one Libyan basin, the samples rule out spore producers that were abundant through the later Ordovician and whose spores reached the rocks as well as later ones did. They cannot rule out rare producers, spores that did not reach those rocks, or plants that lived elsewhere. Samples from one well can share a well effect, but are assumed otherwise independent.
3. **For vascular plants, the answer depends on assumptions the fossils cannot check.** If finds accrued at a steady rate, the fossil record of their bodies is unlikely (95% confidence) to have begun before about 429 Ma, matching the youngest clock; this limits the fossil record, not the date of origin. Counted as fewer independent places, it moves to about 432–434 Ma. If recognisable bodies appeared as soon as the group did, the older clocks' dates (444–452 Ma) would need such fossils to have been found 8 to 12 times less often before the oldest find than after it (2.4 to 4.5 if counted cautiously), or less if they appeared later.
4. **One fossil can move the limit by 20 million years.** Counting one set of tiny spore-capsule fragments from Late Ordovician Oman moves the calculated limit for land-plant bodies back by about 20 million years. Such limits cannot date the delay between plant origins and their fossil bodies.

## What it means

An older, sparsely recorded early flora is *allowed* by the evidence, but not *required* by it. An early lineage can leave few fossils because it was uncommon, local, hard to recognise or seldom preserved. Our calculations show what such conditions would allow, not which clock is right. Clock studies should state these assumptions and carry their uncertainty into the final dates.

## How this work was done

The study was carried out by two AI systems, Claude (Anthropic) and Codex (OpenAI), working as research collaborators with Barry Brook, who set its direction; an independent review by GPT-6 Pro (OpenAI) shaped this revision. The AI systems read the sources, built the models, cross-checked each other and drafted the text. Every number traces to a file and script in the public repository; this summary and the full paper are snapshots of it.

## Sources


Harris, Brogan J., James W. Clark, Dominik Schrempf, Gergely J. Szöllősi, Philip C. J. Donoghue, Alistair M. Hetherington, and Tom A. Williams. 2022. "Divergent Evolutionary Trajectories of Bryophytes and Tracheophytes from a Complex Common Ancestor of Land Plants." *Nature Ecology & Evolution* 6 (11): 1634–43. https://doi.org/10.1038/s41559-022-01885-x.

Hooper, Jack, Annabel M. Worth, Charles H. Wellman, James W. Clark, and Philip C. J. Donoghue. 2026. "Reconstructing Spore Evolution Recalibrates the Timescale of Land Plant Diversification." *Nature Plants*. https://doi.org/10.1038/s41477-026-02432-3.

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